

**LINUX
SCENARIO
BASED Q&A
DAY 7**

Q1. Command works but no man page is available

🧩 Scenario:

You've installed a software package, and while the command executes correctly, running `man <command>` returns:
"No manual entry for <command>"

🧠 Possible Root Cause:

Many Linux distributions separate executable binaries from their documentation. The man pages are often part of a different package (typically named `-doc`).

🔧 Solution:

1. Install the corresponding documentation package:
`sudo apt install <package-name>-doc`

✅ Tip:

Some tools use the `info` system instead of `man`. If the man page is missing, try:
`info <command>`

Q2. New User Created but Can't Login via SSH

🧩 Scenario:

A newly created user exists on the system, but SSH login fails with a "Permission denied" error.

🧠 Possible Root Cause:

This typically happens due to incorrect permissions on the user's home directory or SSH configuration files (e.g., `.ssh` and `authorized_keys`).

🔧 Solution:

1. Check and correct home directory permissions:
`ls -ld /home/username`
`chmod 700 /home/username`
2. Fix SSH-related file permissions:
`chmod 700 ~/.ssh`
`chmod 600 ~/.ssh/authorized_keys`

✅ Tip:

SSH enforces strict permission rules for security. Even minor deviations can prevent login.

Q3. User Gets Logged Out Immediately After Login, Password Expired

🧩 Scenario:

A user logs in via SSH or terminal but is instantly logged out with a message like:
"Your password has expired", and is not prompted to change it.

🧠 Possible Root Cause:

The user's password has expired, but the system isn't allowing an interactive password reset.

 Solution:

1. Reset the user's password manually:
`sudo passwd username`
2. Check password expiration details:
`sudo chage -l username`

 Tip:

You can control password aging policies in `/etc/login.defs` to avoid unexpected lockouts.

Q4. Unable to su to Another User, "Authentication Failure"

 Scenario:

Running `su - username` results in an "Authentication failure" error, even though the password is correct.

 Possible Root Cause:

- The user account is locked.
- The user has an invalid login shell.

 Solution:

1. Unlock the user account:
`sudo passwd -u username`
2. Set a valid shell for the user:
`sudo usermod -s /bin/bash username`

 Tip:

Always ensure that the user has a valid shell listed in `/etc/passwd` (e.g., `/bin/bash`) for successful login.

Q5. sudo Prompts for Password Every Time

 Scenario:

Even though the user is in the sudoers list, sudo always prompts for a password.

 Possible Root Cause:

This is the default behavior of sudo for security reasons.

 Solution:

1. Open the sudoers file safely using visudo:
`sudo visudo`
2. Add the following line to allow passwordless sudo for a specific user:
`username ALL=(ALL) NOPASSWD:ALL`

 Tip:

Use the `NOPASSWD` directive with caution, ideally for trusted users or restrict it to specific commands for better security.

Q6. Unable to Delete User: “User is Currently Logged In”

🧩 Scenario:

Attempting to delete a user with `userdel` results in an error stating the user is currently logged in.

🧠 Possible Root Cause:

The user has an active session (e.g., shell or background process), which blocks deletion.

🔧 Solution:

1. Terminate all active sessions for the user:

```
pkill -u username
```

2. Delete the user and their home directory:

```
sudo userdel -r username
```

✅ Tip:

Use `who`, `w`, or `ps -u username` to check active sessions or processes before attempting deletion.

Q7. `id` Shows Incorrect Groups After Adding User to a New Group

🧩 Scenario:

User has been added to a new group, but `id username` doesn't reflect the change.

🧠 Possible Root Cause:

Group membership is cached in the current session. The update won't apply until a new session starts.

🔧 Solution:

1. Ask the user to log out and log back in to refresh group membership.

2. Alternatively, restart the shell session:

```
exec su - username
```

✅ Tip:

For an immediate switch to a new group within the same session, use:

```
newgrp groupname
```

Q8. New User Created but Home Directory is Missing

🧩 Scenario:

User was created successfully, but `/home/username` directory is not found.

🧠 Possible Root Cause:

The user was created without the `-m` option, or default settings in `/etc/login.defs` were changed to skip home directory creation.

🔧 Solution:

1. Create the directory manually:

```
sudo mkdir /home/username
```

```
sudo chown username:username /home/username
```

2. Or recreate the user with home directory: `sudo useradd -m username`

✅ Tip: Always ensure a home directory is created unless you're setting up a system or service account.

Q9. Users Can't Log In After Editing /etc/passwd

🧩 Scenario:

Manual changes were made to /etc/passwd, and now users are unable to log in.

🧠 Possible Root Cause:

A syntax error or corrupted entry in /etc/passwd can break authentication for all users.

🔧 Solution:

1. Check for syntax issues:

```
pwck
```

2. Restore the backup:

```
sudo cp /etc/passwd- /etc/passwd
```

✅ Tip:

Always use vipw instead of editing /etc/passwd directly, it locks the file and performs basic syntax checks for safety.

Q10. User Account Locked Accidentally

🧩 Scenario:

User is unable to log in and receives an "account locked" message.

🧠 Possible Root Cause:

The account was locked using passwd -l or due to multiple failed login attempts.

🔧 Solution:

1. Unlock the account:

```
sudo passwd -u username
```

✅ Tip:

Check /etc/shadow for accounts marked with ! which indicates the account is locked.